

1. Introduction

Quality control is an important part of manufacturing because it helps detect process problems before they lead to large numbers of defective products. In this project, Statistical Process Control (SPC) tools are used to analyze a DC motor manufacturing process and identify whether the process is stable and capable of producing acceptable products.

The case study focuses on the winding resistance (WR) of DC motors, measured in ohms (Ω). Winding resistance is an important quality characteristic because abnormal resistance values may indicate problems such as poor winding, overheating, material inconsistency, or internal electrical defects.

The dataset includes 30 production batches, with 5 motors inspected in each batch. For each batch, winding resistance measurements and the number of defective units are recorded. The analysis will be carried out both manually and using Minitab, as required.

The main objective of this project is to detect process instability, identify possible causes of defects, and suggest corrective actions to improve the manufacturing process. The tools used include X-bar and R charts, p-chart, Fishbone diagram, Pareto chart, revised control charts, normality test, process capability analysis, and run chart.

2. Data Description

The dataset used in this study represents quality inspection results from a DC motor manufacturing process. A total of 30 batches were analyzed, with 5 motors inspected in each batch (subgroup size $n = 5$).

For each batch, the dataset includes:

- Five winding resistance (WR) measurements (in ohms, Ω), one for each motor
- Number of defective motors (np) identified during inspection

The data therefore consists of raw measurement values and defect counts, which will be used to evaluate process stability and quality performance using statistical tools.

Batch No	WR1 (Ω)	WR2 (Ω)	WR3 (Ω)	WR4 (Ω)	WR5 (Ω)	np Defective
B01	4.410	4.470	4.500	4.550	4.570	0
B02	4.340	4.400	4.440	4.490	4.530	1
B03	4.470	4.520	4.550	4.580	4.630	0
B04	4.420	4.470	4.500	4.550	4.560	0
B05	4.440	4.490	4.530	4.570	4.620	0
B06	4.380	4.430	4.470	4.510	4.510	1
B07	4.440	4.480	4.500	4.530	4.550	0
B08	4.470	4.520	4.570	4.600	4.640	0
B09	4.370	4.430	4.450	4.500	4.550	0
B10	4.500	4.530	4.560	4.600	4.610	1
B11	4.420	4.470	4.500	4.550	4.560	0
B12	4.380	4.430	4.470	4.520	4.550	0
B13	4.650	4.700	4.730	4.760	4.760	2
B14	4.450	4.500	4.520	4.560	4.570	3
B15	4.410	4.470	4.500	4.550	4.570	0
B16	4.330	4.490	4.540	4.580	4.710	1
B17	4.410	4.440	4.470	4.510	4.520	1
B18	4.480	4.520	4.550	4.580	4.620	0
B19	4.550	4.620	4.810	4.630	4.640	3
B20	4.410	4.460	4.480	4.520	4.530	0
B21	4.470	4.520	4.550	4.570	4.590	0
B22	4.350	4.420	4.460	4.520	4.550	1
B23	4.460	4.500	4.530	4.570	4.590	0
B24	4.400	4.460	4.500	4.540	4.550	0
B25	4.430	4.480	4.510	4.540	4.540	1
B26	4.490	4.530	4.560	4.590	4.630	0
B27	4.400	4.460	4.480	4.530	4.530	0
B28	4.450	4.490	4.520	4.560	4.580	0
B29	4.390	4.440	4.460	4.500	4.510	0
B30	4.440	4.490	4.520	4.560	4.590	0

3. Methodology

To analyze the manufacturing process, a structured Statistical Process Control (SPC) approach was followed. The objective is to detect process instability, identify sources of variation, and improve overall product quality.

The analysis was carried out in two parts:

- Manual calculations, to understand the underlying statistical methods
- Minitab analysis, to validate results and visualize process behavior

The workflow followed in this project is summarized in Figure (X).

Step	Tool	Purpose
1	Data Overview	Understand the dataset and spec limits
2	X-bar & R Chart	See the problem — out of control points
3	p-Chart	See how bad the defect rate got
4	Fishbone Diagram	Investigate why — find all possible causes
5	Pareto Chart	Rank the causes — focus on the vital few
6	Revised Control Charts	Apply fixes — show the process improved
7	Normality Test	Validate before running Cpk
8	Cpk Before	Prove the process was not capable
9	Cpk After	Prove the fix worked with numbers
10	Run Chart	Show the full story in one simple chart

The analysis steps are as follows:

- 1. X-bar and R Charts**
Used to evaluate process stability by monitoring the process mean and variation.
- 2. P-Chart**
Used to analyze the proportion of defective units and assess quality performance.
- 3. Root Cause Analysis (Fishbone Diagram)**
Used to identify possible causes of variation and defects.
- 4. Pareto Analysis**
Used to determine the most significant sources of defects.
- 5. Process Improvement**
Corrective actions are proposed based on the identified root causes.
- 6. Revised Control Charts**
Used to verify whether the process becomes stable after improvements.
- 7. Normality Test**
Conducted to validate assumptions required for capability analysis.
- 8. Process Capability Analysis (Cpk)**
Used to evaluate whether the process meets specification requirements.

9. Run Chart

Used to visualize process trends over time.

4. Calculation Procedure

This section explains how the main statistical values were calculated from the raw dataset before creating the control charts. The calculations include the range, average range, subgroup mean, proportion defective, and control limits.

4.1 Range Calculation (R)

$$R = X_{\max} - X_{\min}$$

$$R_{B01} = 4.570 - 4.410 = 0.160$$

Batch No	WR1 (Ω)	WR2 (Ω)	WR3 (Ω)	WR4 (Ω)	WR5 (Ω)	np Defective	Range (Ω)
B03	4.470	4.520	4.550	4.580	4.630	0	0.160
B04	4.420	4.470	4.500	4.550	4.560	0	0.140
B05	4.440	4.490	4.530	4.570	4.620	0	0.180
B06	4.380	4.430	4.470	4.510	4.510	1	0.130
B07	4.440	4.480	4.500	4.530	4.550	0	0.110
B08	4.470	4.520	4.570	4.600	4.640	0	0.170
B09	4.370	4.430	4.450	4.500	4.550	0	0.180
B10	4.500	4.530	4.560	4.600	4.610	1	0.110
B11	4.420	4.470	4.500	4.550	4.560	0	0.140
B12	4.380	4.430	4.470	4.520	4.550	0	0.170
B13	4.650	4.700	4.730	4.760	4.760	2	0.110
B14	4.450	4.500	4.520	4.560	4.570	3	0.120
B15	4.410	4.470	4.500	4.550	4.570	0	0.160
B16	4.330	4.490	4.540	4.580	4.710	1	0.380
B17	4.410	4.440	4.470	4.510	4.520	1	0.110
B18	4.480	4.520	4.550	4.580	4.620	0	0.140
B19	4.550	4.620	4.810	4.630	4.640	3	0.260
B20	4.410	4.460	4.480	4.520	4.530	0	0.120
B21	4.470	4.520	4.550	4.570	4.590	0	0.120
B22	4.350	4.420	4.460	4.520	4.550	1	0.200
B23	4.460	4.500	4.530	4.570	4.590	0	0.130
B24	4.400	4.460	4.500	4.540	4.550	0	0.150
B25	4.430	4.480	4.510	4.540	4.540	1	0.110
B26	4.490	4.530	4.560	4.590	4.630	0	0.140
B27	4.400	4.460	4.480	4.530	4.530	0	0.130
B28	4.450	4.490	4.520	4.560	4.580	0	0.130
B29	4.390	4.440	4.460	4.500	4.510	0	0.120
B30	4.440	4.490	4.520	4.560	4.590	0	0.150

I. Sum of Range (Ω)

$$\sum R = 4.62$$

II. Average Range

$$\bar{R} = \frac{\sum R}{30} = \frac{4.62}{30} = 0.154$$

4.2 Subgroup Mean (X-bar)

$$\bar{X} = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}$$

$$\bar{X}_{B01} = \frac{4.410 + 4.470 + 4.500 + 4.550 + 4.570}{5} = 4.500$$

Batch No	WR1 (Ω)	WR2 (Ω)	WR3 (Ω)	WR4 (Ω)	WR5 (Ω)	np Defective	Range (Ω)	X-bar (Ω)
B01	4.410	4.470	4.500	4.550	4.570	0	0.160	4.500
B02	4.340	4.400	4.440	4.490	4.530	1	0.190	4.440
B03	4.470	4.520	4.550	4.580	4.630	0	0.160	4.550
B04	4.420	4.470	4.500	4.550	4.560	0	0.140	4.500
B05	4.440	4.490	4.530	4.570	4.620	0	0.180	4.530
B06	4.380	4.430	4.470	4.510	4.510	1	0.130	4.460
B07	4.440	4.480	4.500	4.530	4.550	0	0.110	4.500
B08	4.470	4.520	4.570	4.600	4.640	0	0.170	4.560
B09	4.370	4.430	4.450	4.500	4.550	0	0.180	4.460
B10	4.500	4.530	4.560	4.600	4.610	1	0.110	4.560
B11	4.420	4.470	4.500	4.550	4.560	0	0.140	4.500
B12	4.380	4.430	4.470	4.520	4.550	0	0.170	4.470
B13	4.650	4.700	4.730	4.760	4.760	2	0.110	4.720
B14	4.450	4.500	4.520	4.560	4.570	3	0.120	4.520
B15	4.410	4.470	4.500	4.550	4.570	0	0.160	4.500
B16	4.330	4.490	4.540	4.580	4.710	1	0.380	4.530
B17	4.410	4.440	4.470	4.510	4.520	1	0.110	4.470
B18	4.480	4.520	4.550	4.580	4.620	0	0.140	4.550
B19	4.550	4.620	4.810	4.630	4.640	3	0.260	4.650
B20	4.410	4.460	4.480	4.520	4.530	0	0.120	4.480
B21	4.470	4.520	4.550	4.570	4.590	0	0.120	4.540
B22	4.350	4.420	4.460	4.520	4.550	1	0.200	4.460
B23	4.460	4.500	4.530	4.570	4.590	0	0.130	4.530
B24	4.400	4.460	4.500	4.540	4.550	0	0.150	4.490
B25	4.430	4.480	4.510	4.540	4.540	1	0.110	4.500
B26	4.490	4.530	4.560	4.590	4.630	0	0.140	4.560
B27	4.400	4.460	4.480	4.530	4.530	0	0.130	4.480
B28	4.450	4.490	4.520	4.560	4.580	0	0.130	4.520
B29	4.390	4.440	4.460	4.500	4.510	0	0.120	4.460
B30	4.440	4.490	4.520	4.560	4.590	0	0.150	4.520

I. Sum of Subgroup Means

$$\sum \bar{X} = 135.51$$

II. Average of Subgroup Means

$$\bar{\bar{X}} = \frac{135.51}{30} = 4.517$$

4.3 R Chart Control Limits

For subgroup size:

$$n=5$$

From Appendix VI:

$$D_3=0$$

$$D_4=2.114$$

Since:

$$\bar{R} = 0.154$$

The R chart limits are:

$$UCL_R = D_4 \times \bar{R}$$

$$UCL_R = 2.114 \times 0.154 = 0.3256$$

$$CL_R = \bar{R} = 0.154$$

$$LCL_R = D_3 \times \bar{R}$$

$$LCL_R = 0 \times 0.154 = 0$$

4.4 X-bar Chart Control Limits

For the X-bar chart, the control limits are calculated using the constant A_2 . This constant is obtained from Appendix VI according to the subgroup size n .

Since the subgroup size is:

$$n=5$$

From Appendix VI:

$$A_2=0.577$$

Since:

$$\bar{\bar{X}} = 4.517$$

$$\bar{R} = 0.154$$

The X-bar chart limits are:

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \times \bar{R}$$

$$UCL_{\bar{X}} = 4.517 + (0.577 \times 0.154) = 4.6059$$

$$CL_{\bar{X}} = \bar{\bar{X}} = 4.517$$

$$LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \times \bar{R}$$

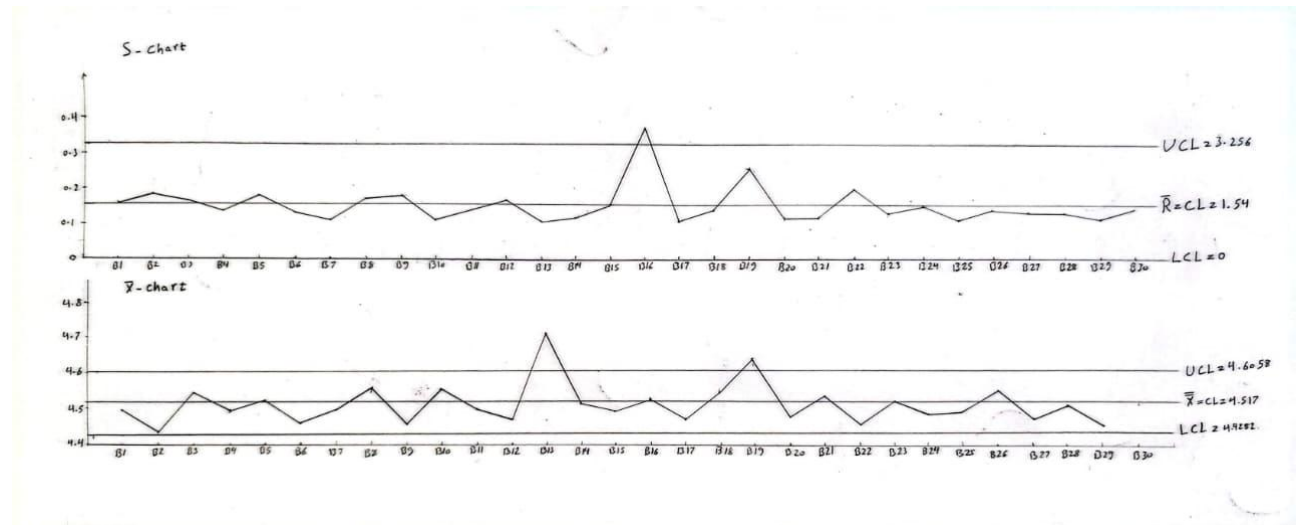
$$LCL_{\bar{X}} = 4.517 - (0.577 \times 0.154) = 4.4281$$

4.5 Control Charts

Using the calculated control limits, the X-bar and R control charts were constructed to evaluate the stability of the manufacturing process.

4.5.1 Manual Control Charts

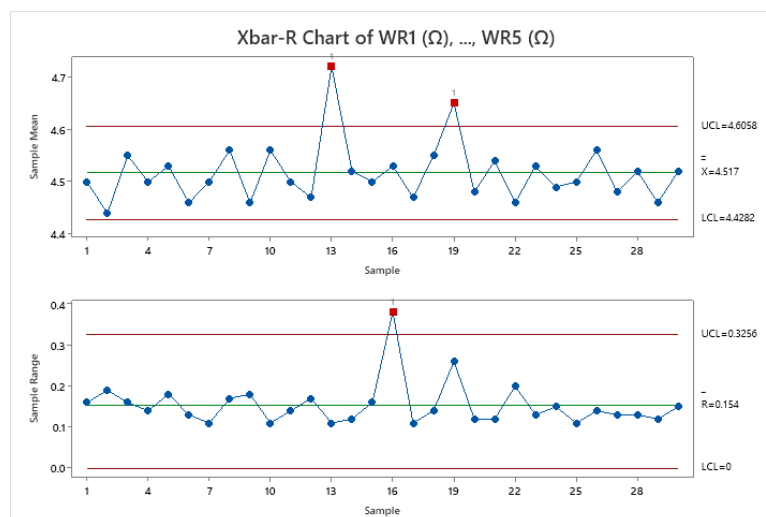
The control charts were first constructed manually using the calculated control limits to evaluate process stability.



- For the **R chart**, the values of R were plotted against the batch number, along with the control limits UCL_R , CL_R , and LCL_R .
- For the **X-bar chart**, the subgroup means $\bar{X}$ were plotted against the batch number, along with the control limits $UCL_{\bar{X}}$, $CL_{\bar{X}}$, and $LCL_{\bar{X}}$.

4.5.2 Minitab Control Charts

The control charts were also generated using Minitab to verify the manual results and provide clearer visualization.



4.6 Control Chart Interpretation

The constructed X-bar and R control charts were analyzed to evaluate the stability of the manufacturing process.

In the R chart, Batch B16 was found to exceed the upper control limit. This indicates the presence of a special cause of variation within the batch, suggesting abnormal process variability.

In the X-bar chart, Batches B13 and B19 were found to exceed the control limits. This indicates that the process mean has shifted, which is a sign of instability in the process.

The presence of out-of-control points in both charts shows that the process is not in statistical control, as both the variation and the mean are affected by special causes.

These results indicate that the manufacturing process is unstable and requires further investigation to identify and eliminate the causes of variation.

5. P-Chart Analysis

To evaluate the quality performance of the process, a p-chart was constructed based on the number of defective units in each batch.

5.1 Proportion Defective Calculation

The proportion defective for each batch is calculated as:

$$p = \frac{np}{n}$$

Where:

np = number of defective motors in the batch

$n=5$ = number of motors inspected in each batch

$$p_{B02} = \frac{1}{5} = 0.20$$

Batch No	WR1 (Ω)	WR2 (Ω)	WR3 (Ω)	WR4 (Ω)	WR5 (Ω)	np Defective	Range (Ω)	X-bar (Ω)	p Defect Rate
B01	4.410	4.470	4.500	4.550	4.570	0	0.160	4.500	0.00
B02	4.340	4.400	4.440	4.490	4.530	1	0.190	4.440	0.20
B03	4.470	4.520	4.550	4.580	4.630	0	0.160	4.550	0.00
B04	4.420	4.470	4.500	4.550	4.560	0	0.140	4.500	0.00
B05	4.440	4.490	4.530	4.570	4.620	0	0.180	4.530	0.00
B06	4.380	4.430	4.470	4.510	4.510	1	0.130	4.460	0.20
B07	4.440	4.480	4.500	4.530	4.550	0	0.110	4.500	0.00
B08	4.470	4.520	4.570	4.600	4.640	0	0.170	4.560	0.00
B09	4.370	4.430	4.450	4.500	4.550	0	0.180	4.460	0.00
B10	4.500	4.530	4.560	4.600	4.610	1	0.110	4.560	0.20
B11	4.420	4.470	4.500	4.550	4.560	0	0.140	4.500	0.00
B12	4.380	4.430	4.470	4.520	4.550	0	0.170	4.470	0.00
B13	4.650	4.700	4.730	4.760	4.760	2	0.110	4.720	0.40
B14	4.450	4.500	4.520	4.560	4.570	3	0.120	4.520	0.60
B15	4.410	4.470	4.500	4.550	4.570	0	0.160	4.500	0.00
B16	4.330	4.490	4.540	4.580	4.710	1	0.380	4.530	0.20
B17	4.410	4.440	4.470	4.510	4.520	1	0.110	4.470	0.20
B18	4.480	4.520	4.550	4.580	4.620	0	0.140	4.550	0.00
B19	4.550	4.620	4.810	4.630	4.640	3	0.260	4.650	0.60
B20	4.410	4.460	4.480	4.520	4.530	0	0.120	4.480	0.00
B21	4.470	4.520	4.550	4.570	4.590	0	0.120	4.540	0.00
B22	4.350	4.420	4.460	4.520	4.550	1	0.200	4.460	0.20
B23	4.460	4.500	4.530	4.570	4.590	0	0.130	4.530	0.00
B24	4.400	4.460	4.500	4.540	4.550	0	0.150	4.490	0.00
B25	4.430	4.480	4.510	4.540	4.540	1	0.110	4.500	0.20
B26	4.490	4.530	4.560	4.590	4.630	0	0.140	4.560	0.00
B27	4.400	4.460	4.480	4.530	4.530	0	0.130	4.480	0.00
B28	4.450	4.490	4.520	4.560	4.580	0	0.130	4.520	0.00
B29	4.390	4.440	4.460	4.500	4.510	0	0.120	4.460	0.00
B30	4.440	4.490	4.520	4.560	4.590	0	0.150	4.520	0.00

5.2 Sum of Defectives

$$\sum np = 15$$

5.3 Average Proportion Defective

$$\sum n = 30 \times 5 = 150$$

$$\bar{p} = \frac{\sum np}{\sum n} = \frac{15}{150} = 0.10$$

5.4 P-Chart Control Limits

Since:

$$\bar{p} = 0.10$$

and the subgroup size is:

$$n=5$$

The p-chart limits are:

$$UCL_p = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$UCL_p = 0.10 + 3\sqrt{\frac{0.10(1 - 0.10)}{5}}$$

$$UCL_p = 0.5025$$

$$CL_p = \bar{p} = 0.10$$

$$LCL_p = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$LCL_p = 0.10 - 3\sqrt{\frac{0.10(1 - 0.10)}{5}}$$

$$LCL_p = -0.3025$$

Since the lower control limit is negative:

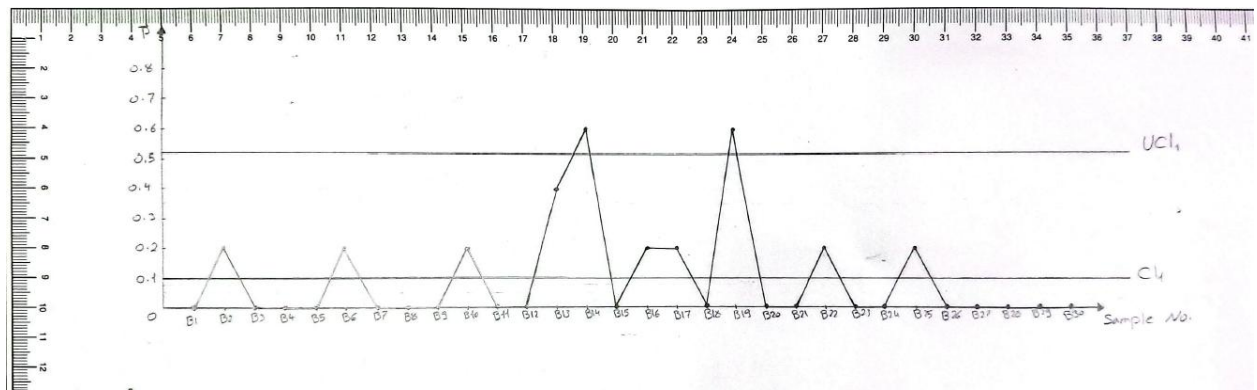
$$LCL_p = 0$$

5.5 Control Charts (P-Chart)

5.5.1 Manual Control Charts

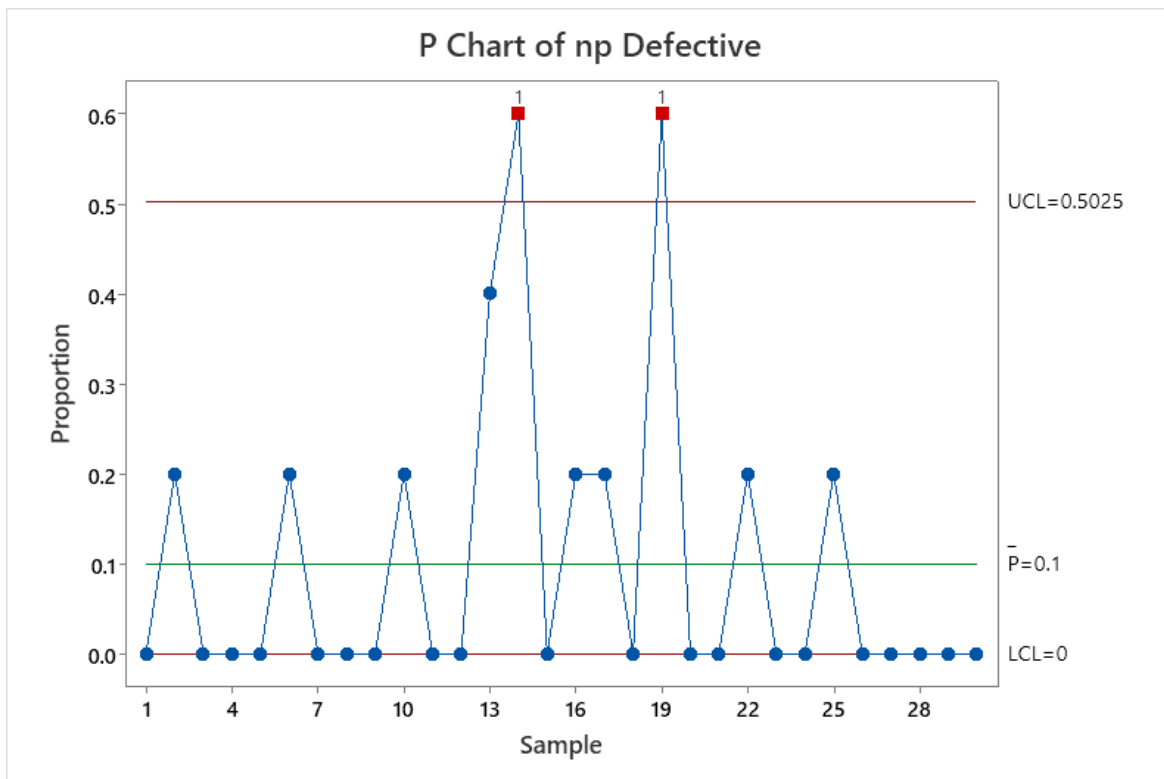
The p-chart was first drawn manually using the calculated values:

The values of p were plotted against the batch number, along with the control limits UCL_p , CL_p , and LCL_p



5.5.2 Minitab Control Charts

The p-chart was also generated using Minitab to verify the manual results and provide clearer visualization.



5.6 P-Chart Interpretation

The p-chart was analyzed to assess the stability of the defect rate

From the chart, Batches B14 and B19 were found to exceed the upper control limit. This indicates that these batches have an abnormally high proportion of defective units, suggesting the presence of special causes affecting product quality.

The remaining batches lie within the control limits, indicating that the defect rate is generally stable except for the identified out-of-control points.

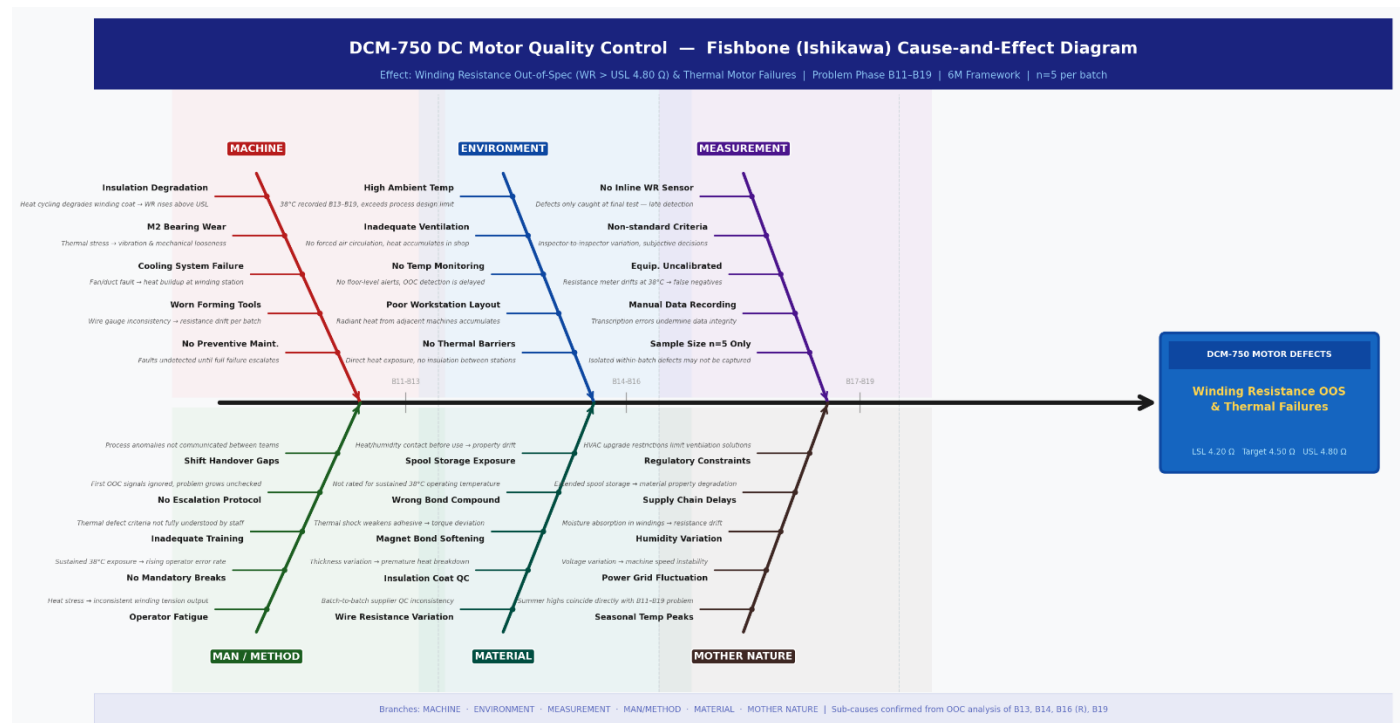
The p-chart confirms that the instability observed in the X-bar and R charts has a direct impact on the defect rate. The presence of out-of-control points indicates that variations in the process lead to an increase in defective products.

6. Root Cause Analysis (Fishbone Diagram)

To investigate the causes of the process instability identified in the control charts, a fishbone (Ishikawa) diagram was developed.

The diagram categorizes potential sources of variation into six main groups: Machine, Method (Man), Material, Measurement, Environment, and Mother Nature.

The analysis focuses on the batches identified as out of control (B13, B14, B16, and B19), where abnormal variation and high defect rates were observed.



6.1 Discussion of Causes

The fishbone diagram highlights several possible causes contributing to the observed process instability.

- Under Machine, issues such as cooling system failure, bearing wear, and lack of preventive maintenance may lead to increased temperature and resistance variation.
- Under Environment, high ambient temperature and poor ventilation can contribute to thermal stress on the motors.
- Under Material, variations in insulation coating and bonding materials may result in inconsistent winding resistance.
- Under Measurement, lack of proper monitoring and calibration may delay the detection of defects.
- Under Man/Method, operator-related issues such as inadequate training and lack of escalation procedures may allow problems to persist.
- Under Mother Nature, external factors such as humidity and seasonal temperature changes may also affect process performance.

These causes provide a basis for further analysis and prioritization using Pareto analysis.

7. Pareto Analysis

Purpose of Pareto Analysis

To identify the most significant sources of defects in the manufacturing process, a Pareto analysis was conducted. This method is based on the principle that a small number of causes are responsible for the majority of problems.

The analysis categorizes defects into key cause groups derived from the root cause (fishbone) analysis.

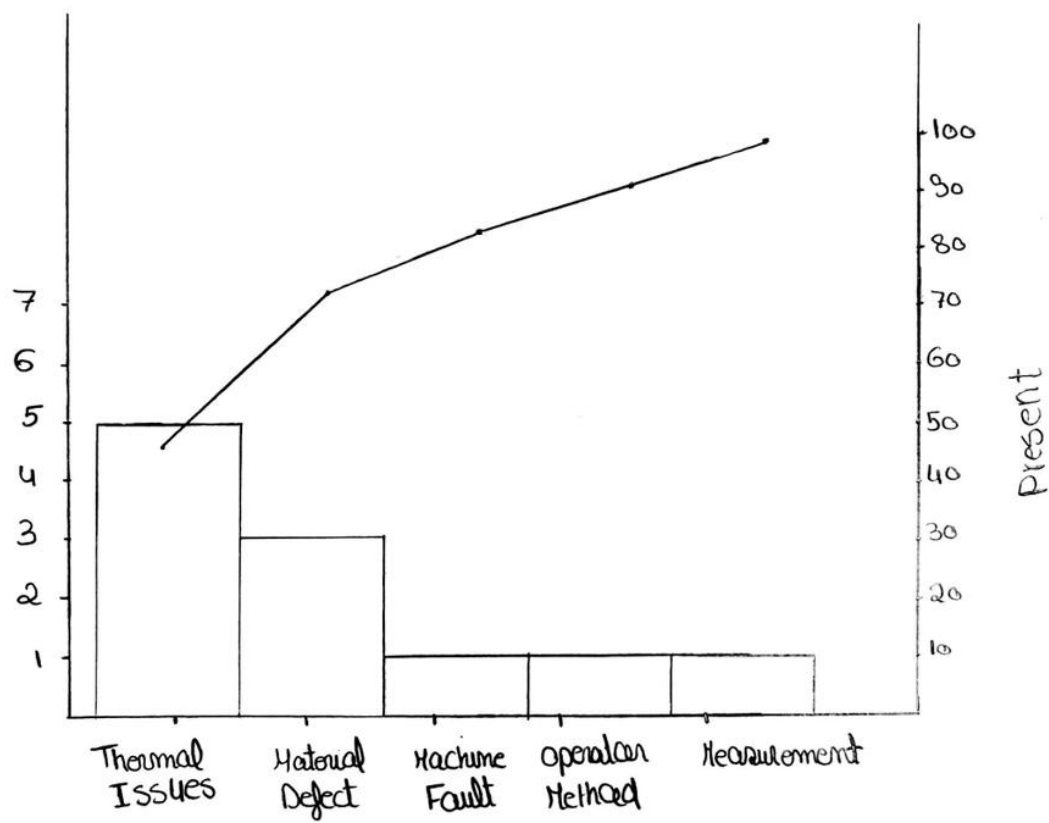
7.1 Pareto Data Table

Before constructing the Pareto chart, the causes of defects were grouped into major categories and their frequencies were counted.

Cause	Count	Percent	Cumulative %
Thermal Issues	5	45.5%	45.5%
Material Defect	3	27.3%	72.7%
Machine Fault	1	9.1%	81.8%
Operator/Method	1	9.1%	90.9%
Measurement	1	9.1%	100.0%

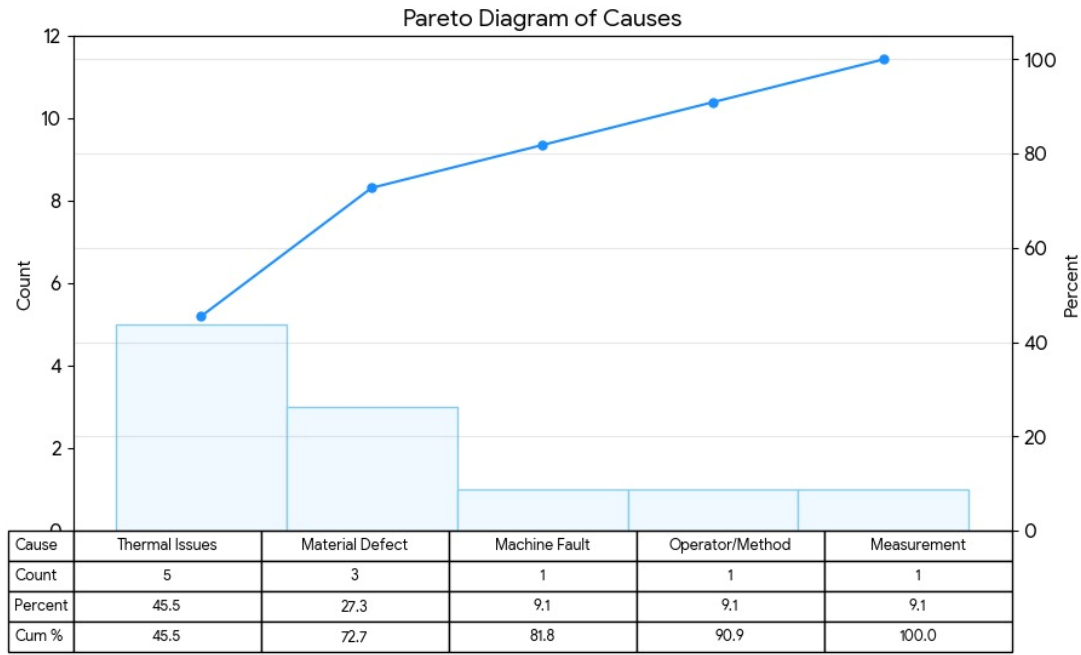
7.2 Manual Pareto Chart

The Pareto chart was first constructed manually by arranging the causes in descending order of frequency and calculating the cumulative percentage.



7.3 Minitab Pareto Chart

The Pareto chart was also generated using Minitab to verify the manual results and provide a clearer visual representation.



7.4 Pareto Interpretation

The Pareto chart shows that thermal-related issues are the largest contributor to defects. Thermal issues account for 45.5% of the total causes, while material defects account for 27.3%.

Together, thermal and material causes represent 72.7% of the total defects. This means that improvement efforts should focus mainly on controlling temperature conditions and improving material quality.

Conclusion

The Pareto analysis shows that most defects come from a small number of major causes. Therefore, focusing on thermal issues and material defects should produce the greatest improvement in the manufacturing process.

8. Process Improvement Actions

Based on the control chart analysis, fishbone diagram, and Pareto analysis, the main causes of process instability were identified as **thermal-related issues and material**

inconsistencies, along with minor contributions from machine and operational factors.

To address these issues, corrective actions were proposed and implemented as follows:

8.1 Thermal Control Improvements

- Installation of improved **cooling systems** to reduce motor overheating
- Enhancement of **workshop ventilation** to maintain stable ambient temperature
- Implementation of **temperature monitoring systems** during production

8.2 Material Quality Control

- Standardization of **insulation coating processes**
- Improved quality inspection of **raw materials**
- Supplier quality verification to reduce batch-to-batch variation

8.3 Equipment Maintenance

- Regular **preventive maintenance schedules** introduced
- Inspection and replacement of worn components such as **bearings**
- Calibration of equipment to ensure consistent operation

8.4 Process and Operator Improvements

- Training programs for operators to improve handling and inspection
- Introduction of **clear escalation procedures** for abnormal conditions
- Reduction of manual errors through better process control

Conclusion

These corrective actions target the primary sources of defects identified in the analysis. The implementation of these improvements is expected to reduce process variation and improve product quality.

9. Revised Control Charts

Following the implementation of the proposed improvement actions, the control charts were recalculated to evaluate the stability of the process under improved conditions.

9.1 Revised R Chart

Batch B16 was identified as an out-of-control point in the R chart and was excluded for recalculating the control limits.

The revised average range is:

$$\sum R_{\text{revised}} = 4.62 - 0.380 = 4.24$$
$$\bar{R}_{\text{revised}} = \frac{4.24}{29} = 0.146$$

From Appendix VI, for n=5

$$D_3 = 0, \quad D_4 = 2.114$$
$$UCL_R = 2.114 \times 0.146 = 0.309$$
$$CL_R = 0.146$$
$$LCL_R = 0$$

9.2 Revised X-bar Chart

Sum of Revised Subgroup Means

$$\sum \bar{X}_{\text{revised}} = 135.51 - 4.530 = 130.98$$
$$\bar{\bar{X}}_{\text{revised}} = \frac{130.98}{29} = 4.5166$$

Using the revised value of $\bar{R}$, the X-bar chart limits are recalculated as:

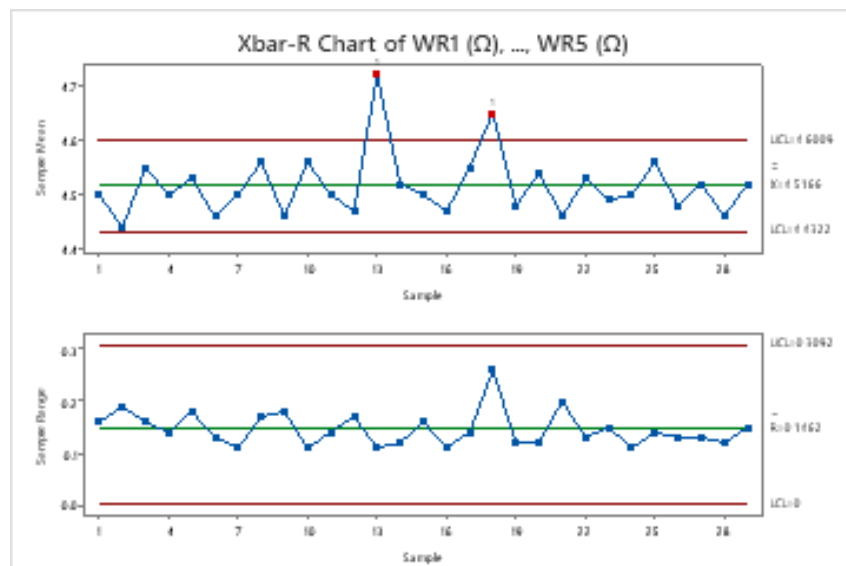
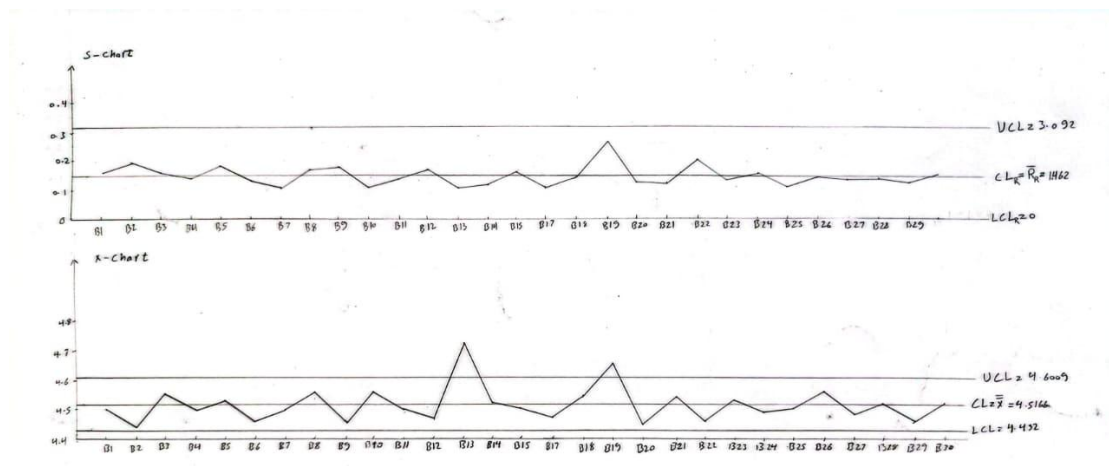
$$UCL_{\bar{X}} = 4.5166 + (0.577 \times 0.1462) = 4.6009$$

$$CL_{\bar{X}} = 4.5166$$

$$LCL_{\bar{X}} = 4.5166 - (0.577 \times 0.1462) = 4.4322$$

9.3 Revised Control Charts (Manual and Minitab)

The revised control charts were constructed manually using the recalculated control limits and verified using Minitab to evaluate the process after removing the special-cause variation.



9.4 Interpretation

The revised R chart shows that all points lie within the control limits, indicating that the process variation is now stable after removing the special-cause batch.

However, the revised X-bar chart still shows that Batches B13 and B19 exceed the control limits. This indicates that although within-batch variation has been controlled, the process mean is still affected by special causes.

These remaining out-of-control points are likely related to systematic factors identified earlier, especially thermal conditions and material quality. Therefore, additional corrective actions are required to fully stabilize the process.

This reflects that process improvement is iterative, and multiple corrective actions may be needed to achieve full statistical control.

9.5 Additional Corrective Actions

The revised control charts indicated that although process variation was stabilized, the process mean remained out of control due to Batches B13 and B19.

From the previous analysis (Fishbone and Pareto), these out-of-control points were associated with:

- Thermal issues (overheating during production)
- Material inconsistencies (insulation and resistance variation)

To address these issues, additional corrective actions were implemented:

- Improved temperature control and cooling systems
- Better monitoring of thermal conditions during production
- Standardization of material quality and supplier control

After implementing these actions, the abnormal conditions causing B13 and B19 were eliminated.

Therefore, these batches were considered to be affected by assignable causes and were excluded from the final dataset.

9.6 Final X-bar Chart

The subgroup means were recalculated after excluding B16, B13, and B19:

$$\sum \bar{X}_{final} = 130.98 - 4.720 - 4.650 = 121.61$$

$$\bar{\bar{X}}_{final} = \frac{121.61}{27} = 4.504$$

Using:

$$A_2 = 0.577$$

$$\bar{R}_{revised} = 0.146$$

$$UCL_{\bar{X}} = 4.504 + (0.577 \times 0.146) = 4.588$$

$$CL_{\bar{X}} = 4.504$$

$$LCL_{\bar{X}} = 4.504 - (0.577 \times 0.146) = 4.420$$

9.7 Final Control Charts (Manual and Minitab)

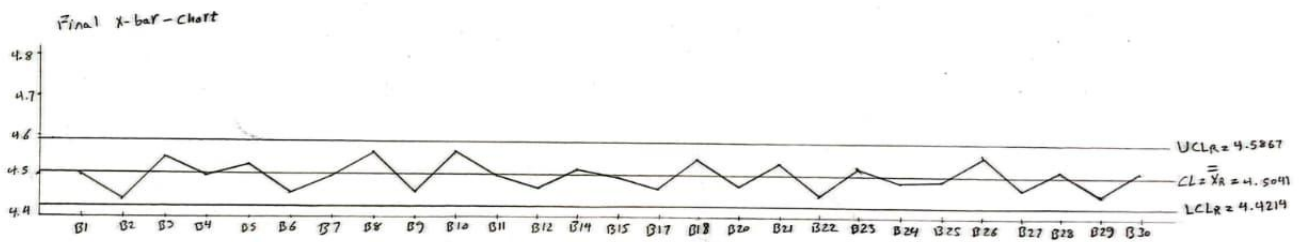
The final X-bar chart was constructed using the recalculated control limits after excluding B16, B13, and B19 due to identified assignable causes:

Since the R chart was already stable, only the X-bar chart was redrawn to assess the stability of the process mean after implementing the corrective actions.

9.7.1 Manual Control Chart

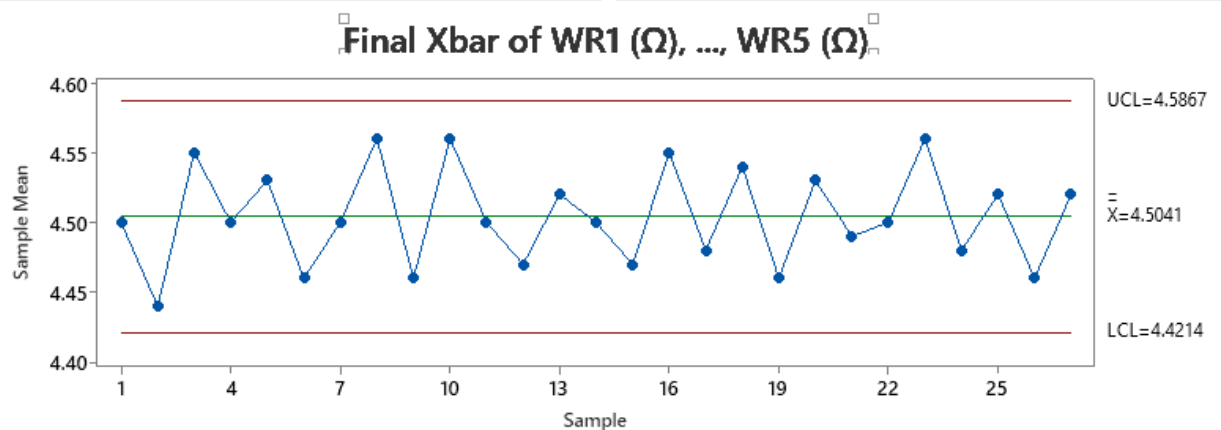
The final X-bar chart was first constructed manually using the calculated control limits.

- The subgroup means $\bar{X}$ were plotted against the batch number
- The control limits $UCL_{\bar{X}}$, $CL_{\bar{X}}$, $LCL_{\bar{X}}$ were drawn as horizontal reference lines



9.7.2 Minitab Control Chart

The final X-bar chart was also generated using Minitab to verify the manual results and provide a clearer visualization.



9.8 Final Interpretation

The final X-bar chart shows that all subgroup means lie within the control limits, indicating that the process mean is now stable after implementing the additional corrective actions.

Combined with the previously stabilized R chart, this confirms that the manufacturing process is now in statistical control.

This confirms that the implemented corrective actions successfully addressed both process variation and process mean shifts, resulting in a fully stable manufacturing process.

9.9 Revised P-Chart

After implementing the corrective actions and removing batches affected by assignable causes (B13, B14, B16, and B19), the defect rate was recalculated to evaluate the process quality under improved conditions.

The revised total number of defectives is:

$$\sum np_{revised} = 15 - (2 + 3 + 1 + 3) = 6$$

The total number of inspected units becomes:

$$\sum n = 26 \times 5 = 130$$

The revised average proportion defective is:

$$\bar{p}_{revised} = \frac{6}{130} = 0.0462$$

9.9.1 Revised P-Chart Control Limits

$$UCL_p = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$UCL_p = 0.0462 + 3\sqrt{\frac{0.0462(1 - 0.0462)}{5}} \approx 0.328$$

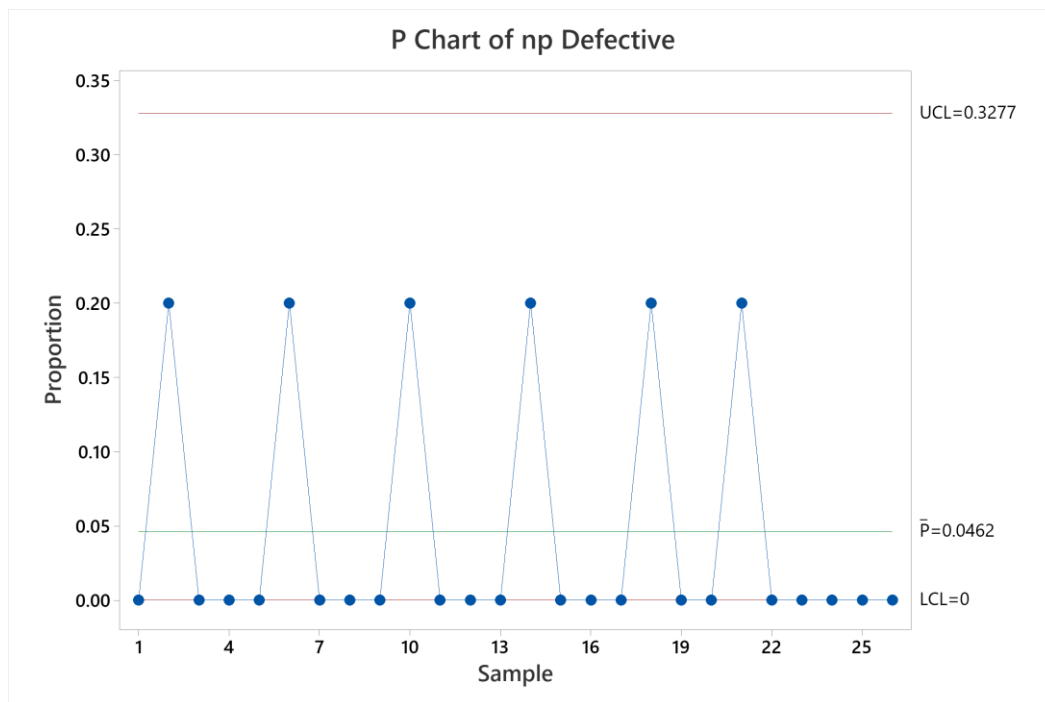
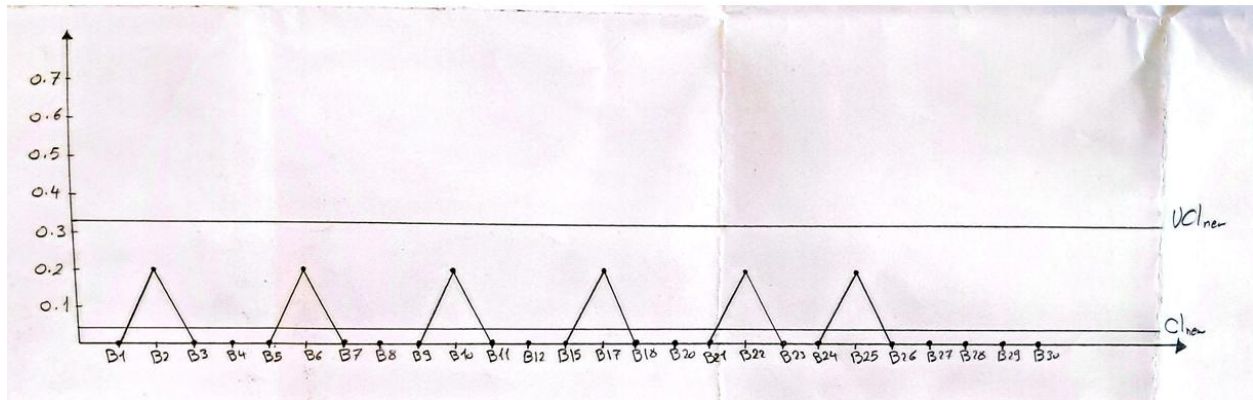
$$CL_p = \bar{p} = 0.0462$$

$$LCL_p = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$LCL_p = 0.0462 - 3\sqrt{\frac{0.0462(1 - 0.0462)}{5}} = -0.235 \rightarrow 0$$

9.9.2 Revised Control Charts (Manual and Minitab)

The revised p-chart was constructed manually using the recalculated values and verified using Minitab to evaluate the defect rate after removing the special-cause variation.



9.9.3 Interpretation

The revised p-chart shows that all sample proportions lie within the control limits, indicating that the defect rate has significantly improved after implementing the corrective actions.

This confirms that the root causes identified through the Fishbone and Pareto analyses were successfully addressed, resulting in improved product quality.

This confirms that the process improvement successfully reduced the defect rate and achieved stable product quality.

10. Normality Test

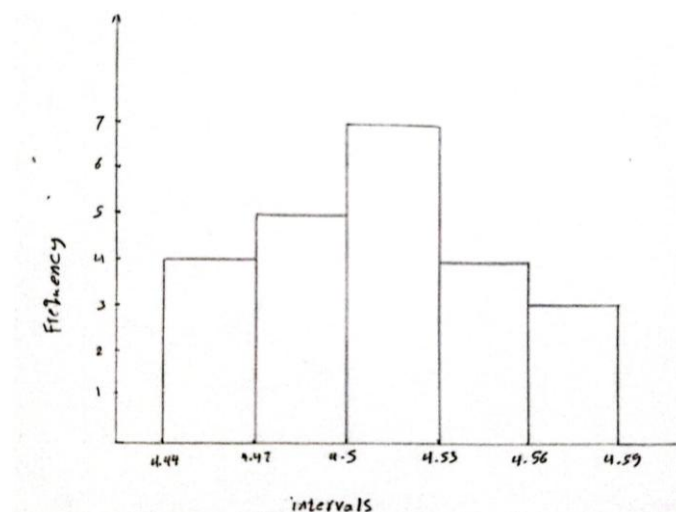
The normality of the data was evaluated to verify whether the assumption of normal distribution is satisfied before performing process capability analysis.

10.1 Manual Normality Check

The normality of the data was assessed manually using a histogram of the subgroup means $\bar{X}$. The values were grouped into intervals, and the frequency of each interval was plotted to observe the shape of the distribution.

The frequency distribution table used to construct the histogram is shown below.

Interval (Ω)	Frequency
4.44 – <4.47	4
4.47 – <4.50	5
4.50 – <4.53	7
4.53 – <4.56	4
4.56 – <4.59	3
Total	23

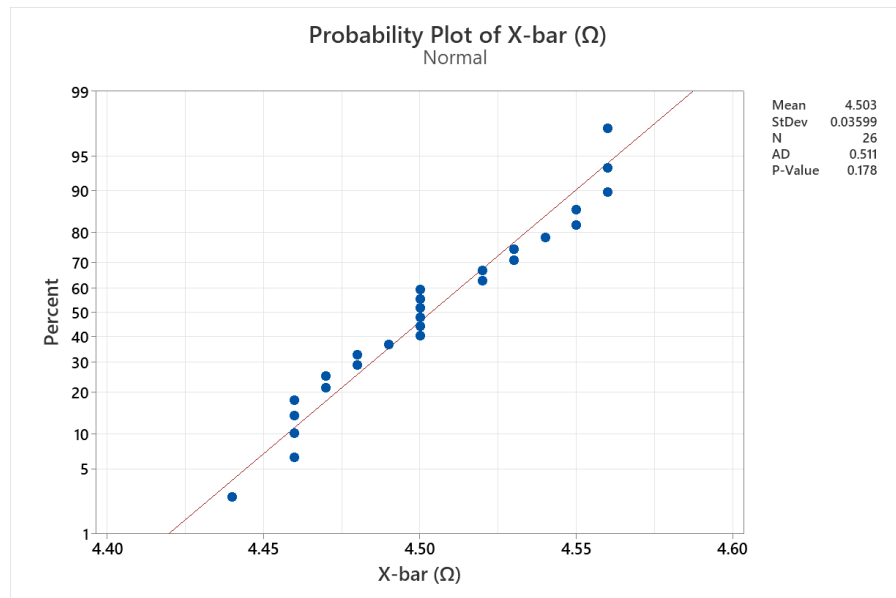


Interpretation

The histogram shows an approximately symmetric, bell-shaped distribution, indicating that the data is approximately normally distributed.

10.2 Minitab Normality Test

The normality of the data was further evaluated using Minitab with the Anderson-Darling test.



Interpretation

The normal probability plot shows that the data points closely follow the reference line, indicating approximate normality.

The p-value obtained from the Anderson-Darling test is 0.178, which is greater than 0.05. Therefore, the assumption of normality is satisfied.

11. Process Capability Analysis (Cpk)

The process capability analysis was conducted to evaluate the ability of the manufacturing process to produce products within the specified limits after implementing the corrective actions.

11.1 Specification Limits

From the project specifications:

$$LSL=4.20$$

$$USL=4.80$$

11.2 Process Mean and Standard Deviation

From the final revised dataset:

$$\bar{\bar{X}} = 4.50346$$

$$\bar{R} = 0.146$$

For subgroup size:

$$n=5$$

From Appendix VI:

$$d_2=2.326$$

The process standard deviation is calculated as:

$$\sigma = \frac{\bar{R}}{d_2}$$
$$\sigma = \frac{0.146}{2.326} = 0.0628$$

11.3 Process Capability Index (Cpk)

The process capability indices are calculated as follows:

$$C_{pu} = \frac{USL - \bar{\bar{X}}}{3\sigma}$$

$$C_{pu} = \frac{4.80 - 4.50346}{3(0.0628)} = 1.57$$

$$C_{pl} = \frac{\bar{\bar{X}} - LSL}{3\sigma}$$

$$C_{pl} = \frac{4.50346 - 4.20}{3(0.0628)} = 1.61$$

$$C_{pk} = \min(C_{pu}, C_{pl})$$

$$C_{pk} = \min(1.57, 1.61) = 1.57$$

11.4 Interpretation

The calculated value of $C_{pk} = 1.57$ indicates that the process is capable of producing products within the specified limits.

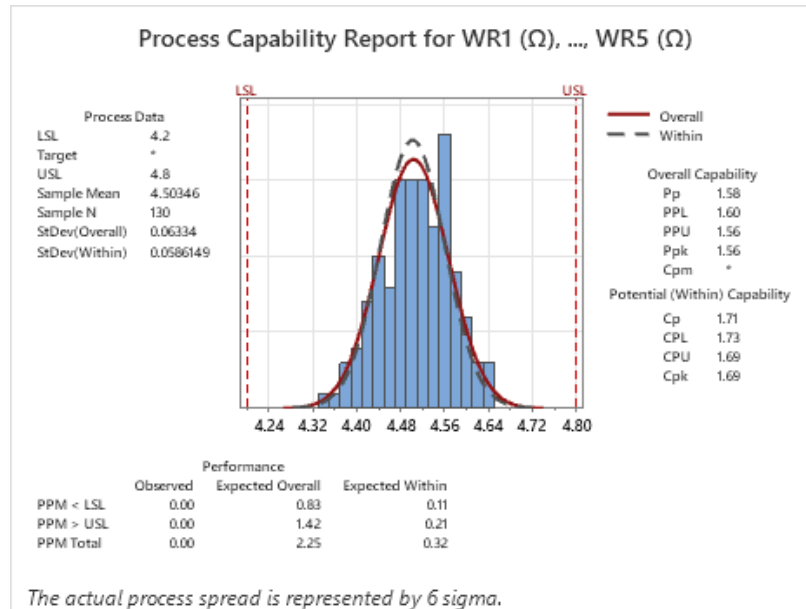
Since $C_{pk} > 1.33$, the process is considered highly capable and well-centered within the specification range.

This confirms that the corrective actions implemented during the analysis were effective in improving both process stability and product quality.

Compared to the initial process condition, the process capability has significantly improved after eliminating the assignable causes of variation.

11.5 Minitab Process Capability Analysis

The process capability was further evaluated using Minitab based on the raw measurement data.



Interpretation

The Minitab output shows that the process capability index C_{pk} is approximately 1.69, which is higher than the manually calculated value.

Since $C_{pk} > 1.33$, the process is considered highly capable of meeting the specification limits.

The histogram indicates that the process data is well-centered between the lower and upper specification limits, with minimal variation.

These results confirm that the process is both stable and capable after implementing the corrective actions.

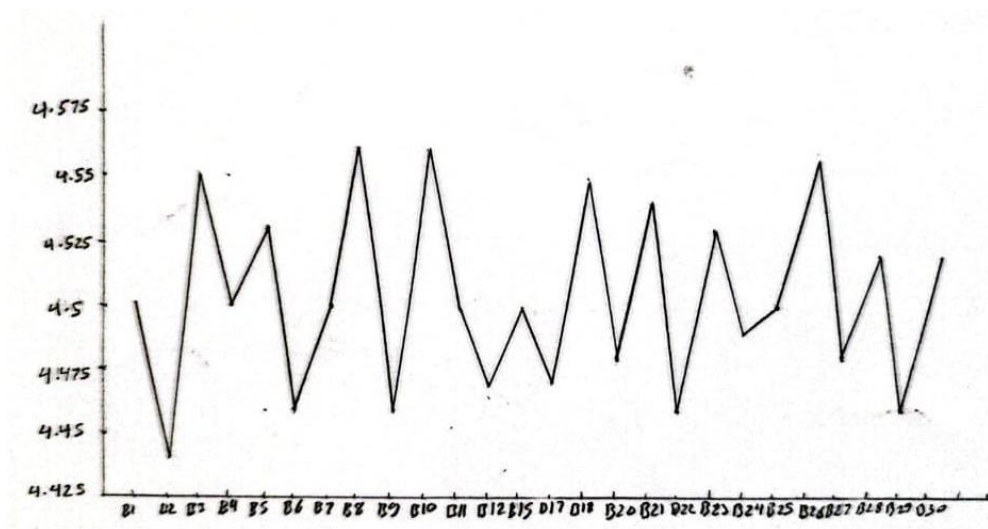
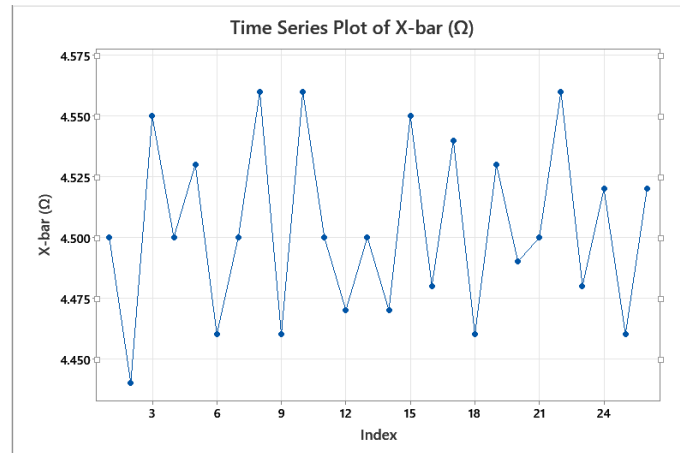
12. Run Chart

12.1 Run Chart Construction

A run chart was constructed to visualize the behavior of the process over time and to detect any trends, shifts, or patterns in the data.

The subgroup means $\bar{X}$ were plotted against the batch number in chronological order.

Although the run chart uses the same subgroup means as the X-bar chart, it focuses only on identifying patterns over time without the use of control limits.



12.2 Run Chart Interpretation

The run chart shows that the subgroup means fluctuate randomly around the process mean without any systematic patterns.

No significant upward or downward trends are observed, and the data points do not exhibit any prolonged shifts.

This indicates that the process behavior is stable over time after implementing the corrective actions.

The results of the run chart are consistent with the $\bar{X}$ -bar and R chart analysis, further confirming that the process is operating under statistical control.

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13. Final Conclusion

This study applied Statistical Process Control (SPC) techniques to evaluate and improve the quality of the DC motor manufacturing process based on winding resistance (WR) measurements.

The initial analysis using X-bar and R control charts revealed that the process was not in statistical control, with several out-of-control points indicating the presence of assignable causes of variation. The p-chart analysis further confirmed instability in the defect rate, highlighting inconsistencies in product quality.

Root cause analysis using the Fishbone (Ishikawa) diagram and Pareto chart identified the major sources of variation, particularly thermal issues and material inconsistencies. Based on these findings, corrective actions were implemented to address these problems.

Following the implementation of corrective actions, the revised control charts showed a significant improvement in process stability. The R chart confirmed that process variation was controlled, while the final X-bar chart indicated that the process mean was stable and within control limits.

The normality test verified that the data follows an approximately normal distribution, satisfying the assumptions required for process capability analysis.

The process capability analysis demonstrated that the process is capable of meeting the specification limits, with a manually calculated $C_{pk} = 1.57$ and a Minitab result of approximately $C_{pk} = 1.69$. These values indicate that the process is highly capable and well-centered within the specification range.

The run chart further confirmed that the process is stable over time, with no significant trends or shifts observed after the corrective actions.

Overall, the results show that the implemented improvements successfully enhanced both process stability and product quality. The manufacturing process is now operating under statistical control and is capable of consistently producing products within the required specifications.